AKSHATHA SRIKANTHA

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EDUCATION

University of California, Irvine | *Research and Teaching Assistant* **Expected - 2025**
MSCS - Master of Computer Science **GPA 3.9**
Coursework: Operating System, Database and Data Management, Algorithms, Data Structures, Machine Learning.

MVJ College of Engineering, Bangalore **Aug 2017 – Jun 2021**
BE in Computer Science **GPA 3.82**
Coursework: Discrete Mathematical Structures, Unix Programming, Microprocessors, Computer Organization, Computer Networks, OS, IoT, ML, AI, Big Data Analytics, OOPS

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL, C++, Java, HTML/CSS, MongoDB, PostgreSQL, Redis
Tools & API: Git, Docker, Apache Kafka, Power BI, AWS, REST APIs
Frameworks & Libraries: React, Vue.js, Node.js, Flask, PySpark, scikit-learn, Pandas, TensorFlow, OpenCV
ML & AI: NLP, Computer Vision, Financial ML, Sentiment Analysis, Explainable AI, Streamlit

EXPERIENCE

Research Assistant – Anatomy and Neurobiology Lab, UC Irvine — Apr 2025 – Present

- Conduct experiments under Dr. Kei Igarashi to investigate entorhinal hippocampal circuit dysfunction in Alzheimer's disease.
- Perform neural signal processing of EEG signals with Python and NeuraLynx to analyze memory related activity.

IBM- Data Engineer - Metlife Client — Oct 2021 – Jun 2024

- Built and optimized actuarial data pipelines using PySpark and Python, reducing risk analysis report turnaround by 80%.
- Designed scalable MongoDB data models and applied advanced SQL validation to resolve 100+ data quality issues across systems.
- Created Power BI dashboards to visualize actuarial KPIs, improving decision making speed for business stakeholders.
- Implemented Apache Kafka on Linux/Ubuntu for real time data streaming, enabling event driven actuarial risk monitoring.
- Partnered with actuaries and analysts to automate regulatory and financial reporting, streamlining quarterly workflows.

PUBLICATIONS

- Developed a system to classify emotional stress from EEG signals using SVM and Genetic Algorithm, with noise filtering and feature extraction. Presented findings at the 2021 ICA ECA Conference and published in IEEE.

PROJECTS

Distilled Reasoner – Tiny LLM Distillation Framework *(PyTorch, PEFT, FastAPI)* **Oct 2025**
- Architected a model distillation framework to transfer complex reasoning capabilities from large teacher models (GPT-4) to a compact, efficient 1.3B parameter student model, enabling faster and cheaper inference.
- Synthesized a high quality dataset of instruction answer pairs using GPT-4 on reasoning benchmarks (GSM8K, ARC) and fine tuned base models (LLaMA-3-1B) using PEFT-LoRA for parameter efficient learning.
- Deployed the distilled model with a vLLM runtime and FastAPI wrapper, achieving high throughput, low latency inference with streaming token support for production ready serving.

LLM Alignment & Evaluation Suite *(Hugging Face, Streamlit, Perspective API)* **Aug 2025**
- Engineered an automated evaluation pipeline to benchmark the reliability and safety of open source LLMs (LLaMA-3, Mistral) against metrics for truthfulness, toxicity, and bias.
- Integrated datasets like TruthfulQA and RealToxicityPrompts, leveraging the Hugging Face Inference API and Perspective API to run evaluations and calculate multi dimensional scores.
- Built an interactive Streamlit dashboard to visualize comparative model performance through radar charts and metric plots, providing clear insights into model strengths and ethical limitations.

AI Debugger – Self-Correcting Code Reasoning Agent *(LangChain, FastAPI, Mistral-7B)* **Jun 2025**
- Developed an autonomous debugging agent using a LangChain powered Mistral-7B model to analyze Python error traces, suggest intelligent fixes, and iteratively correct code in a sandboxed environment.
- Designed a secure backend with FastAPI to execute user submitted code, capture errors, and orchestrate the agent's reasoning loop until a valid solution was found or the process converged.
- Demonstrated capabilities in agentic problem solving and automated testing, simulating a software engineer's debugging process to enhance code quality and reliability.

Dream Recorder & Analyzer (EEG Sleep Analysis Pipeline) - Startup Project **Jun 2025 - present**
- Developed an end to end EEG analysis pipeline using MNE and YASA in Google Colab
- Automated REM detection, bandpower computation, and sleep efficiency metrics across .edf files.
- Integrated Google Drive support with optimized preprocessing and visualizations, exporting structured results to CSV.